
Soundgaze 2.0: Evaluating Dimensionality Reduction Methods for Interactive 3D Audio Similarity Exploration

Lester Martin
lester.martin@emory.edu

Parker Wischhover
parker.wischhover@emory.edu

Abstract

Music streaming services largely focus on recommendation: predicting what a listener might want to hear next. This project takes a complementary view, treating a music collection as a space to be explored rather than filtered. Soundgaze 2.0 is an interactive 3D system that maps 8,000 songs from the Free Music Archive into a navigable point cloud where proximity reflects sonic similarity. Two challenges shape the design. Audio embeddings often fail to match human perceptual judgments, and compressing those embeddings to 3D for visualization introduces additional distortion. To address the first, a hybrid 578-dimensional representation is constructed from CLAP semantic embeddings combined with hand-crafted acoustic features, and validated against the Timbremetrics and Covers80 perceptual benchmarks. To address the second, three dimensionality reduction methods (PCA, t-SNE, and UMAP) are directly compared on the same hybrid embedding using both structural metrics (Trustworthiness, KNN Recall) and perceptual alignment metrics. Results show t-SNE preserves local neighborhood structure best (Trustworthiness 0.977, KNN Recall 0.224), while PCA performs worst across every metric. All three methods lose substantial perceptual fidelity once compressed to 3D. This finding motivates a system design where nearest-neighbor queries run on the full hybrid space, while the chosen reduction method governs only the visual layout.

1 Introduction

Most modern music tools answer one question: *what should I listen to next?* Streaming services build elaborate recommendation systems that surface tracks based on listening history, collaborative filtering, or content-based similarity. A different and less commonly asked question is *what is in this collection, and how does it relate?* This is the question of exploration rather than recommendation, and it is the motivation for the work presented here.

Building an exploration tool requires a notion of similarity that listeners actually agree with. Modern audio embedding models produce high-dimensional vectors that capture genre, mood, and timbral character, but distances in those spaces do not always reflect listener perception (1). A second challenge follows. Even if the embedding space is perceptually meaningful, projecting hundreds of dimensions into the three coordinates needed for an interactive 3D visualization introduces inevitable distortion. The choice of dimensionality reduction method therefore directly affects whether the resulting layout feels meaningful when the user navigates it.

This paper presents Soundgaze 2.0, an interactive 3D music exploration system built on the Free Music Archive Small dataset (FMA Small). The work tackles both challenges in sequence. First, a hybrid embedding combines pretrained CLAP audio representations with hand-crafted acoustic descriptors, then validates the resulting space against perceptual benchmarks. Second, three dimensionality

reduction methods (PCA, t-SNE, and UMAP) are compared head-to-head on this embedding using structural and perceptual metrics. Third, the analysis informs the design of a web application where 3D layouts serve as navigation aids while similarity queries continue to operate on the full embedding space.

The headline findings are that the hybrid embedding consistently outperforms pure CLAP on every perceptual benchmark tested, that t-SNE preserves local neighborhood structure better than UMAP or PCA at 3D, and that no method can fully retain the perceptual quality of the high-dimensional space. The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the problem. Section 4 describes the technical pipeline. Section 5 presents experimental results. Section 6 describes the web application. Section 7 concludes.

2 Related Work

Content-based music similarity has a long history in music information retrieval. Schedl et al. surveyed the field and noted a recurring tension: audio features that group tracks well algorithmically do not always group them in ways listeners find natural (1). This perceptual gap motivates the choice in this work to combine learned and hand-crafted features rather than relying on either alone.

For visualizing high-dimensional data, three methods dominate the literature. Principal Component Analysis is a classical linear projection that maximizes preserved variance. Van der Maaten and Hinton introduced t-SNE, a nonlinear method that emphasizes preservation of local pairwise similarities and was shown to produce visually clearer cluster structure than PCA on image and document datasets (2). McInnes et al. proposed UMAP, which uses a fuzzy simplicial set representation to preserve both local and global structure while scaling more efficiently than t-SNE (3). Each method has been applied widely in audio contexts, but a direct comparison on a perceptually validated audio similarity space is uncommon.

Two lines of work on audio representation also matter here. Hand-crafted features such as MFCCs, chroma vectors, and spectral descriptors remain competitive baselines for genre clustering and similarity tasks; Ferraro et al. show they can rival learned embeddings on some retrieval tasks (4). CLAP (Contrastive Language-Audio Pretraining) by Wu et al. produced general-purpose audio embeddings via large-scale contrastive training on audio-text pairs and serves as the learned component of the hybrid embedding used here (5). The FMA Small benchmark used as the music corpus was introduced by Defferrard et al. (6).

For evaluation, Trustworthiness from Venna and Kaski measures how faithfully a low-dimensional projection preserves local neighborhood structure, and is a standard metric for assessing manifold preservation (7).

The contribution of this work is a direct benchmark of PCA, t-SNE, and UMAP applied to a perceptually validated hybrid audio embedding, paired with an interactive web tool that puts each reduction in front of the user for qualitative comparison.

3 Problem Formulation

3.1 Setup

Let $\mathcal{T}$ denote a collection of N tracks. For each track $t \in \mathcal{T}$, two embeddings are computed: a 512-dimensional CLAP embedding $c_t \in \mathbb{R}^{512}$ that captures semantic and timbral content, and a 66-dimensional acoustic feature vector $a_t \in \mathbb{R}^{66}$ derived from MFCCs, chroma, spectral descriptors, and rhythmic features. Both vectors are L2-normalized so that downstream cosine similarity behaves as a pure angular comparison.

3.2 Hybrid Embedding and Similarity

The hybrid embedding $h_t \in \mathbb{R}^{578}$ is formed by concatenating the two normalized vectors with magnitude weighting α :

$$h_t = \left[\sqrt{\alpha} c_t \mid \sqrt{(1 - \alpha)} a_t \right]. \quad (1)$$

This construction has a useful property. Standard cosine similarity on h exactly equals a convex combination of the two component similarities:

$$\text{sim}(A, B) = h_A \cdot h_B = \alpha \cdot \cos(c_A, c_B) + (1 - \alpha) \cdot \cos(a_A, a_B). \quad (2)$$

Importantly, no custom distance function is required, and any downstream tool that operates on cosine similarity works directly on the hybrid representation.

3.3 Dimensionality Reduction Objective

For interactive 3D visualization, a function $f : \mathbb{R}^{578} \rightarrow \mathbb{R}^3$ is required that compresses the hybrid space into three coordinates while preserving as much of the relevant structure as possible. Define $\tilde{h}_t = f(h_t)$. Three candidate functions are evaluated: PCA, t-SNE, and UMAP.

The structure to preserve is neighborhood structure. Let $\mathcal{N}_k(t)$ be the set of k nearest neighbors of track t under cosine distance in the hybrid (578-D) space, and let $\tilde{\mathcal{N}}_k(t)$ be the analogous set for $\tilde{h}_t$ under Euclidean distance in 3D. KNN Recall at rank k is the average overlap between these two sets:

$$\text{Recall@}k = \frac{1}{N} \sum_{t \in \mathcal{T}} \frac{|\mathcal{N}_k(t) \cap \tilde{\mathcal{N}}_k(t)|}{k}. \quad (3)$$

Trustworthiness, introduced by Venna and Kaski (7), penalizes the introduction of false neighbors in the projection:

$$T(k) = 1 - \frac{2}{Nk(2N - 3k - 1)} \sum_{t=1}^N \sum_{j \in \tilde{\mathcal{N}}_k(t) \setminus \mathcal{N}_k(t)} (r(t, j) - k), \quad (4)$$

where $r(t, j)$ is the rank of j among the neighbors of t in the hybrid space. A score of 1 means no false neighbors are introduced.

3.4 Challenges

There are two distinct sources of distortion. The first is the gap between any audio embedding and listener perception: a similarity metric defined on h may still rank tracks in ways human listeners would not. This is addressed during embedding construction by combining CLAP with acoustic features and validating the result against perceptual benchmarks (Section 5). The second is the unavoidable information loss when compressing 578 dimensions to 3, which is the central object of study in the dimensionality reduction analysis.

4 Technical Design

4.1 Dataset

Soundgaze 2.0 uses the FMA Small split, which contains 8,000 tracks balanced across 8 genres: Hip-Hop, Pop, Folk, Experimental, Rock, International, Electronic, and Instrumental (6). Each track is a 30-second MP3 clip sampled at 44.1 kHz. Genre and metadata labels come from a multi-level CSV header file shipped with the dataset. For embedding extraction, audio is resampled to 48 kHz for CLAP and 22,050 Hz for acoustic features.

4.2 CLAP Embeddings

Pretrained CLAP embeddings are computed using the LAION-CLAP implementation with the HTSAT-tiny audio backbone (5). Each 30-second clip yields a single 512-dimensional vector capturing semantic content learned from large-scale audio-text contrastive pretraining. CLAP performs well at distinguishing genre, mood, instrumentation, and broad timbral character (for example, separating a “warm” from a “bright” sound). It is less reliable on explicitly rhythmic features such as tempo, or harmonic features such as key and mode, since these were not directly supervised during training.

4.3 Acoustic Embeddings

To complement CLAP, a 66-dimensional vector of hand-crafted acoustic features is extracted for each track using the Python `librosa` library. The breakdown is:

- **MFCCs (40D):** 20 mean and 20 standard-deviation Mel-Frequency Cepstral Coefficients, approximating human perception of timbre.
- **Chroma CQT (12D):** mean chroma per pitch class, capturing the tonal and harmonic profile of the track.
- **Spectral Contrast (7D):** energy difference between spectral peaks and valleys across 7 frequency subbands, a proxy for spectral flatness.
- **Other features (7D):** additional descriptors covering energy level, tempo, onset density, and characteristics related to noisiness and melodic content.

Features are z-score standardized using collection-wide statistics, and per-group weights are applied during normalization (MFCCs are slightly down-weighted while density-related features are slightly up-weighted) before final L2 normalization.

4.4 Hybrid Embedding Construction

The CLAP and acoustic vectors are L2-normalized, scaled by $\sqrt{0.5}$, and concatenated to form a 578-dimensional hybrid embedding (Section 3). The choice of $\alpha = 0.5$ was made empirically: across the perceptual benchmarks reported in Section 5, $\alpha = 0.5$ produced the best alignment between the hybrid similarity ranking and human perceptual rankings.

4.5 Dimensionality Reduction

Three reduction methods are applied to the 578-dimensional hybrid embedding to produce 3D coordinates suitable for rendering. Each is run with a fixed random seed (42) for reproducibility, and the output is min-max normalized per axis to $[0, 1]$ for consistent rendering on the frontend.

- **PCA.** Standard principal component analysis with 3 components, via `sklearn.decomposition.PCA`. A linear baseline that preserves directions of maximum variance but cannot capture nonlinear structure.
- **t-SNE.** Stochastic Neighbor Embedding with 3 components, perplexity 30, and cosine distance, via `sklearn.manifold.TSNE`. The method optimizes a loss that emphasizes preservation of local pairwise similarities.
- **UMAP.** Uniform Manifold Approximation and Projection with 3 components and cosine metric, via `umap-learn`. UMAP attempts to preserve both local and global structure through a fuzzy graph representation.

4.6 Nearest-Neighbor Design Decision

A central design choice in Soundgaze 2.0 is that nearest-neighbor queries operate on the full 578-dimensional hybrid embedding regardless of which reduction method is currently active in the visualization. Switching between PCA, t-SNE, and UMAP changes only the visual layout; the set of tracks returned as “nearest neighbors” for a given query is identical across methods. Section 5 justifies this empirically: the 3D projected neighbors agree with the high-dimensional neighbors only weakly (KNN Recall of 0.22 at best), so using the 3D space for retrieval would degrade similarity quality. A brute-force `sklearn.NearestNeighbors` index over the hybrid embeddings serves all queries; at 8,000 tracks, this returns results in a few milliseconds.

5 Experimental Analysis

Two questions drive the experimental analysis. Does the hybrid embedding actually capture perceptual similarity better than CLAP alone? And how much of that perceptual quality survives projection into 3D under each reduction method?

5.1 Sonic Similarity Verification

To assess whether the hybrid embedding tracks human perception, two external benchmarks are used: Timbremetrics and Covers80.

Timbremetrics. A Python library that compares model-predicted similarity rankings against human perceptual ratings collected across 21 published psychoacoustic studies, reporting standard correlation metrics over pairwise and triplet judgments. Table 1 compares pure CLAP against the hybrid embedding at $\alpha = 0.5$.

Table 1: Timbremetrics comparison: pure CLAP vs. hybrid embedding ($\alpha = 0.5$). The hybrid improves alignment with human perceptual judgments on every metric; the relative Spearman gain of +7.7% is the most informative.

Metric	CLAP (Pure)	CLAP + Acoustic ($\alpha = 0.5$)	Δ
Spearman ρ	0.4455	0.4796	+0.0341
Kendall τ	0.3442	0.3741	+0.0299
Triplet Agreement	0.7232	0.7399	+0.0167
NDCG	0.9110	0.9175	+0.0065
MAE	0.1693	0.1673	-0.0020

The hybrid wins across the board. Combining learned semantic features with hand-crafted acoustic features produces rankings that better match listener judgments than CLAP alone.

Covers80. An 80-pair cover-song dataset (160 tracks total) where each original is paired with a cover. Cover versions share melody and harmony but differ in instrumentation and production, so a perceptually grounded similarity metric should rank the cover among the top neighbors of the original. Table 2 reports retrieval performance.

Table 2: Covers80 retrieval performance. The hybrid improves cover identification across all metrics. Absolute scores are low because cover identification is hard: the perceptual link between an original and its cover often involves abstract melody and harmony rather than direct acoustic similarity.

Metric	CLAP (Pure)	CLAP + Acoustic ($\alpha = 0.5$)	Δ
Top-1	4.5%	6.5%	+2.0
Top-5	14.9%	16.2%	+1.3
MRR	0.1118	0.1291	+0.0173

A small qualitative listening test was also performed. After the hybrid space was computed, 20 tracks from the FMA Small data set were selected and their top neighbor under the hybrid metric retrieved. Each pair of tracks were judged by the authors on their sonic similarity. While anecdotal rather than a formal study, the example aligned with the direction of the quantitative gains in Tables 1 and 2.

These benchmarks established the hybrid 578-dimensional space as a more reasonable proxy for perceptual similarity than raw CLAP embeddings, and it is treated as the ground-truth space for the dimensionality reduction analysis that follows.

5.2 Dimensionality Reduction Evaluation

The three reduction methods are evaluated on two axes: structural preservation (how faithfully the 3D layout reproduces the high-dimensional neighborhood structure) and perceptual preservation (how well the 3D distances continue to match perceptual benchmarks). Table 3 reports the full results.

Three findings stand out.

First, t-SNE preserves local neighborhood structure best, with the highest Trustworthiness (0.9766) and KNN Recall (0.2241) among the three methods. This matches the original t-SNE motivation (2), which emphasizes preservation of local pairwise structure at the cost of global layout fidelity.

Table 3: Dimensionality reduction evaluation against the 578D hybrid ground truth. Structural metrics (Trustworthiness, KNN Recall) measure neighborhood preservation; perceptual metrics (Spearman, Kendall, Triplet, Covers Top- k) compare 3D distances against external benchmarks. t-SNE preserves local structure best; PCA is consistently weakest; all 3D methods lose substantial perceptual fidelity. t-SNE distances are not globally meaningful, so its values on the global Timbremetrics correlations are reported as n/a.

Metric	CLAP Hybrid (578D)	UMAP (3D)	t-SNE (3D)	PCA (3D)
KNN Recall	n/a	0.1258	0.2241	0.0284
Trustworthiness	n/a	0.9524	0.9766	0.8725
Spearman ρ	0.4796	0.1765	n/a	0.2206
Kendall τ	0.3741	0.1287	n/a	0.1659
Triplet Agreement	0.7399	0.5874	n/a	0.6048
Covers Top-1	6.5%	1.3%	3.9%	2.6%
Covers Top-5	17.5%	6.5%	8.4%	6.5%

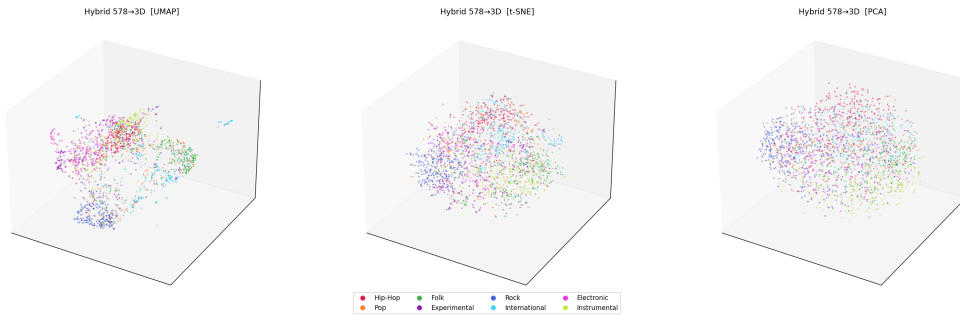


Figure 1: 3D projections of the 578-dimensional hybrid embedding under UMAP (left), t-SNE (middle), and PCA (right). Points are colored by genre. t-SNE produces visually clearer cluster structure consistent with its higher KNN Recall and Trustworthiness; PCA produces a diffuse cloud with weak genre separation.

Second, PCA performs worst on both structural metrics (KNN Recall 0.0284, Trustworthiness 0.8725) and on most perceptual metrics. Audio embeddings, especially when combined with acoustic descriptors, sit on a nonlinear manifold that PCA’s linear projection cannot capture.

Third, every 3D method loses substantial perceptual fidelity relative to the hybrid space. Spearman correlation falls from 0.4796 in 578D to 0.1765 (UMAP), 0.2206 (PCA), or undefined (t-SNE, where global distances are not meaningful). This is unsurprising in absolute terms: collapsing 578 axes into 3 must throw something away. But it has a concrete design implication. Similarity queries in Soundgaze 2.0 should never run on the projected coordinates if perceptual quality matters; they run on the full hybrid embedding instead.

Figure 1 shows the three projections side by side, colored by genre. Visual inspection mirrors the quantitative findings. t-SNE produces tight, identifiable clusters; UMAP shows partial cluster structure with some bridging between genres; PCA produces a diffuse cloud with weak separation.

6 Web Application

The system delivering Soundgaze 2.0 is a full-stack web application that lets users navigate the 8,000-track point cloud in real time, switch between the three reduction methods on demand, and inspect any track's nearest neighbors with audio metadata. A deployed version with 200 songs can be accessed here: <https://soundgaze2.vercel.app/>

6.1 Architecture

The backend is implemented in Python using FastAPI with Uvicorn. On startup, it lazily generates and caches CLAP embeddings, acoustic features, the hybrid 578D matrix, and the three precomputed 3D reductions. Once cached on disk as Parquet and NumPy files, subsequent restarts are fast. A scikit-learn brute-force NearestNeighbors index over the hybrid embedding serves all similarity queries. Three endpoints are exposed:

- GET `/health`: status and currently loaded methods.
- GET `/points?method={pca|tsne|umap}`: returns all 8,000 tracks with their 3D coordinates for the selected method.
- GET `/similar?track_id=N&method=M&k=10`: returns the top- k neighbors of a query track in the hybrid space, paired with their 3D coordinates under the requested visualization.

The frontend is a Next.js 14 application written in TypeScript, with React Three Fiber and Three.js rendering the 3D point cloud. Each track is drawn as a circular sprite, and the selected track is additionally rendered as a 3D sphere for visual prominence. Camera controls support pan, orbit, and zoom; auto-rotation runs at a low rate when the user is idle and pauses on interaction. The styling uses a neo-brutalist aesthetic with the Syne typeface.

6.2 User Experience

The user clicks any point in the cloud to select it. The selected track's 10 nearest neighbors are highlighted in orange, while the selected track itself is highlighted in red. A right-side panel displays the track's title, artist, genre, and a ranked list of neighbors with their similarity scores. Clicking a neighbor in the panel makes it the new selection, supporting iterative exploration through the collection.

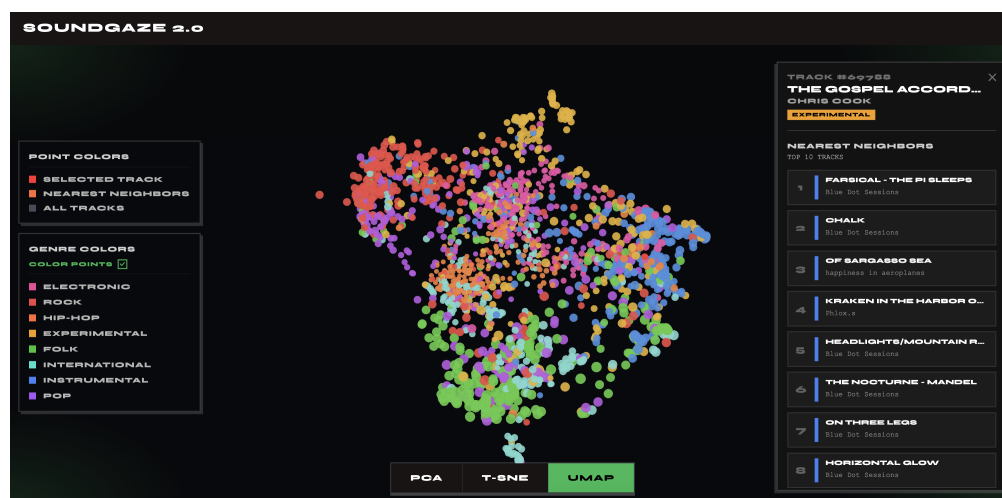


Figure 2: The Soundgaze 2.0 web interface.

A method-switcher widget at the bottom of the canvas toggles between PCA, t-SNE, and UMAP. To make switching feel instantaneous, the frontend prefetches all three coordinate arrays in parallel on initial load. Subsequent switches read from local cache and trigger only a smooth tween between point positions, with no network round-trip required.

6.3 Design Implication of the DR Findings

The k-NN-on-full-hybrid decision from Section 4.6 is what makes the method switcher coherent. Because the neighbor set is identical across visualization methods, switching layout reveals different geometric arrangements of the same neighborhood. A user who finds a cluster of similar tracks in t-SNE space can switch to UMAP and observe how those same tracks redistribute under a different manifold assumption. The reduction comparison is therefore not just an academic exercise: it is a first-class element of the user-facing tool.

7 Conclusion and Future Directions

This work presented Soundgaze 2.0, an interactive 3D system for exploring sonic similarity in a music collection, and used the system as a vehicle for evaluating dimensionality reduction methods on audio embeddings. The contribution is twofold. A 578-dimensional hybrid embedding combining CLAP semantic features with hand-crafted acoustic descriptors was shown to align better with human perceptual judgments than CLAP alone across the Timbremetrics and Covers80 benchmarks. PCA, t-SNE, and UMAP were then directly compared on this embedding, and t-SNE was found to preserve local neighborhood structure most faithfully at 3D, while all three methods lost substantial perceptual fidelity. The empirical result motivated a system design where the 3D projection is purely a visualization aid: nearest-neighbor queries run on the full hybrid embedding to keep similarity quality high.

Several extensions are natural. The first is a more thorough hyperparameter sweep, particularly for t-SNE perplexity and UMAP’s `n_neighbors` and `min_dist`, which were left near defaults in this work. The second is replacing hand-crafted acoustic features with learned representations specifically tuned for rhythmic and harmonic content, which CLAP under-emphasizes. The third is a structured human evaluation: while a listening test interface was built into the web application and an informal pairwise listening confirmation was conducted, a larger study with multiple evaluators would provide more reliable perceptual ground truth than the automated benchmarks alone. Finally, scaling beyond FMA Small to FMA Medium or Large would test whether the structural and perceptual findings hold at larger corpus sizes where t-SNE in particular may not scale gracefully.

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